

**Lab Notebook**  
**Aerial M**  
**ACS SEED - summer 202**

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## Reading summary

### Instruction

For each reading/video, write the title, post the link below it, and write a summary of their content in less than 10 sentences. For example, answer these 3 questions

- What is it about? Or what is the current problem in the covered topic?
- Why is it important to learn about this topic?
- (if applicable) How can we explore this topic or address the problems mentioned above?

### Layers of the GI Tract

<https://youtu.be/7rFrnnrus7Y?si=i4rK0IMfBM6JOPGZ>

Date: May 27, 2025

Summary: *The gastrointestinal tract (GI tract) is a muscular tube that travels through the body from the mouth to the anus, carrying food which we process, resulting in us extracting nutrients in our blood and then eliminating substances that we cannot use.*

Layers: Lumen(hollow of the organ)

- 1.)Mucosa(innermost portion)
- 2.)Submucosa (deep) -thicker area of loose connective tissue
- 3.)muscularis Externa (the outer periphery of the GI tract)-inner layer is (circular coat) - outer layer (longitudinal coat)

4.) Serosa or Adventitia (encased by a sheet of connective tissue)

Mucosa as 3 parts:

- 1.) Stratified squamous- mouth to upper esophagus  
-anus
- 2.) Simpler columnar
- 3.) -stomach to rectum

## Small intestine and food absorption

★ The invisible Universe of the human microbiome ★

<https://youtu.be/5DTrENdWvvM>

Date: May 28, 2025

Summary: The human microbiome are all of our genes outnumbering 100 to 1 microbial genes. Most of the microorganisms aren't bad, they help keep our body in health. We get them from our mother when being born and/or breastfed. Educating our immune cells. When becoming adults, microbes are the first line of defense fighting every germs and protecting their turf (our body) for our health. Different species of microbes have adapted to each habitat they live in whether it's wet, dry, or oily.

## What is the Gut microbiome?

Date: May 28, 2025

Summary: The most important and complex is in the gut, the most diverse fighting off infections, revving up and dampening down our immune systems. They help regulate how much energy we burn and how much fat we store.

(If not functioning right it can lead to diseases, colon cancer, colitis, diabetes, and obesity.)

Microbes are essential to our health and happiness.

# What is Bacteria?

[\(OLD VIDEO\) Bacteria: The Good, The Bad, The Kinda Gross](#)

Instruction: in complete sentences( more than 7 sentences), summarize this video by describing the following

- What is this video about?This video is about the bacteria life
- Highlight important ideas about bacteria, such as where you can find them, what they do to us (good and bad ones), and what antibiotics do to them?
- Date: May 28, 2025
- Summary:
- **Bacteria:** Strep throat: caused by a bacterium streptococcus, which can make you very sick, producing toxins in your body that makes your body have to fight them off.

Bacteria plays a major role in tooth decay, destroying or defuncting the good cells on the tooth barrier. Bacteria have no membrane and no nucleus like in organelle cells.

## Antibiotics:

Most forms of Antibiotics are going to target the structure of prokaryote cells. Most of them are also a broad spectrum, they're not super selective. They can leave you feeling a little sick, or losing your appetite effectively. They destroy the bad bacteria but also the good bacteria.

## Good Bacteria:

Fact: you have more bacterial cells in your body than you have human cells. It is especially found in your digestive system which helps break down the food you eat. They help you synthesise certain molecules and keep other microorganisms in control. Bacteria are also one of the most important decomposers in food webs.

# How to streak agar plate?

Answer these after training in lab with Uyen

1. Describe the experiment you did yesterday by streaking glycerol stocks on agar plate. Provide details so that a person reading this can follow your instructions (protocol).

Once we pipet the bacteria onto the plate we used a sterilized glass hook to spread in the shape of the L up wards and side wards, then moved on the next side and did it four times before entering the incubator.

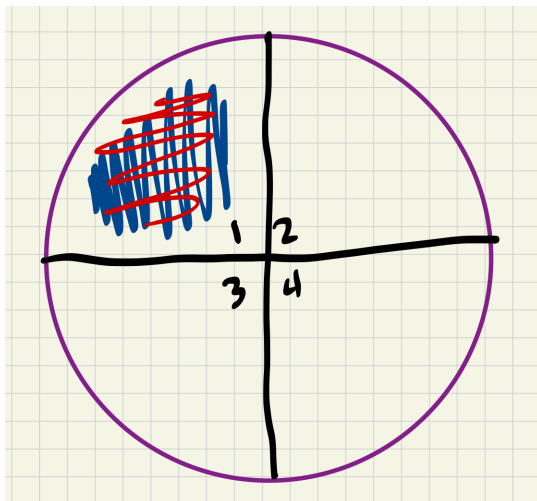
2. Use an app you like, on your ipad, draw an illustration to demonstrate how to do this experiment. ( how to streak plate, how to grow starter culture). See pictures of lab notebook

Here are Softwares that are free, popular, and have many tutorial on Youtube:

- Krita <https://krita.org/en/> – to paint, etc. with drawing tablet
- Inkscape <https://inkscape.org/> – design, draw vector (just need mouse to design)
- Blender <https://www.blender.org/> – design 3D, draw 2D in 3D environment (can use mouse only, but if you have drawing tablet, it allows you to draw more complicated art)

Paste the illustration here:

1. How to streak plate



the two streaks are NOT touching the black nor the edges because those bacteria wont be counted

2. How to grow starter culture

## Serial Dilution

Review this techniques from the following sources and answer the following questions

<https://microbeonline.com/serial-dilution-method/>

<https://www.youtube.com/watch?v=yYWFX4lXc5Y>

1. What is the purpose of serial dilutions?
2. Which step did we use serial dilution in our yogurt experiment? What is the purpose of this serial dilution?
3. Briefly describe how we did serial dilution in our yogurt experiment. Include details such as what yogurt brand we used, what the volume of the serial dilution (yogurt and sterile water)

## Prepare presentation for Final report (Zastrow group meeting and ACS SEED symposium)

1. Slide 1: title: leave it blank for now, add your name, affiliation, and date
2. Slide 2:
  - a. Title: <try to come up with a title after you write content, I will give feedback later>
  - b. Content: <list answers to the below questions as bullet points in your slide>
    - i. What is gut microbiome?

- ii. What is probiotics?
  - iii. Why is it important to do research about them? (hint: how probiotics improve human health?)
  - iv. Which food have probiotics?
- c. Illustration: google to find high resolution images to demonstrate the content discussed above. Post the website link where you find this image on the slide.
- 3. Slide 3
  - a. Title: <try to come up with a title after you write content, I will give feedback later>
  - b. Content:
    - i. What is dietary metals?
    - ii. What is the important role of some important dietary metals? (hint: see this video  
[https://www.youtube.com/watch?v=uZousR\\_FfEE](https://www.youtube.com/watch?v=uZousR_FfEE) )
  - c. Illustration: find images of food that is rich in metals mentioned in above video)
- 4. Slide 4:
  - a. Title: <experiment>
  - b. Content:
    - i. use Biorender <link> to draw a scheme to demonstrate Yogurt expt.
    - ii. Tutorial:
- 5. Slide 5: streak plate
  - a. Draw scheme to demonstrate: done
  - b. Result: we just streaked plate 7/15/2024 and wait till next day to take picture.
- 6. Slide 6: Microscope
  - a. We will add this later, after Wed, when you learn how to prepare your own microscope slides.
- 7. Slide 7: Measure growth curve
  - a. How to measure bacterial growth by monitoring OD600:
    - i. watch this video,  
<https://www.youtube.com/watch?v=rmLqEfaTa9k>
    - ii. Use biorender to draw scheme demonstrate how OD600 works, hint: draw scheme based on Figure 1 from this site.  
<https://byonoy.com/journal/od600-4facts-4tips-bacterial-culture-growth/>
  - b. What is growth curve
    - i. Illustration: use Fig 1 in this link. Paste this link in slide since we borrow their image.  
<https://app.jove.com/v/10511/growth-curves-generating-growth->

[curves-using-colony-forming-units-and-optical-density-measurements?trialstart=1](#)

- ii. In 2-3 bullet points: explain what is growth curve, how and why we monitor it. Hint: watch this  
[https://www.youtube.com/watch?v=JpnH\\_6kqvdw](#)